MIL-C-46507A (MU)
15 October 1965
SUPERSEDING
MIL-C-46507 (ORD)
20 March 1961

MILITARY SPECIFICATION
CASE, CARTRIDGE, 105MM, M14B1

1. SCOPE

1.1 This specification covers Case, Cartridge, 105MM, M14B1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification For
- MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification For

STANDARDS
Federal
- Federal Test Method Standard No. 151 - Metals; Test Methods

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
- MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS
Munitions Command
- D7548025 - Case, Cartridge, 105MM, M14B1
- F7548235 - Cartons, Ammunition for Interplant Shipment of Steel Cartridge Cases, 37MM to 120MM, Inclusive

FSC 1315

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2.1 (Cont.) DRAWINGS 'Munitions Comd.'

IEL-7548025 - Index to Inspection Equipment List for Case, Cartridge, 105MM, M14B1

PUBLICATIONS
QAI-14-210-1 - Quality Assurance Instructions, Preparation of Ammunition Data Cards

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required (see 6.1).

3.3 Material.- The supplier shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Mechanical properties.- The mechanical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each section.

3.4.1 Thermal treatment.- The entire cartridge case shall be subjected to a thermal treatment after the heading operation by heating above the upper critical temperature point, quenching and tempering to the desired properties.

3.5 Stress relief.- The cartridge case shall be adequately stress relieved at the times and temperatures suitable for development of desired properties.

3.6 Protective coating.- Protective coating shall be as specified on the applicable drawing and referenced specifications and shall apply to all surfaces of the cartridge case regardless of contour configuration.

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3.7 Maximum profile.- Maximum profile of the cartridge case shall comply with the requirements of the applicable drawing.

3.8 Functional.- The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase

Cold Temperature Phase

Excess, Excess Pressure Phase (for information only)

3.8.1 Obturation.- The cartridge case shall effectively seal the breech of the weapon against gases generated during firing.

3.8.2 Soundness.- After firing, the cartridge case shall be free of splits or cracks, except as permitted by Table II.

3.8.3 Extraction.- After firing, the cartridge case shall be extracted by the service extractors when the weapon is operated in the normal manner (see 6.3).

3.8.4 Primer retention.- The cartridge case shall retain its primer during firing and extraction.

3.9 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.9.1.1 Flow of metal.- The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.9.2 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.9.3 Foreign matter.- No cartridge case shall contain chips, dirt, oil, grease, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Initial production sample.- Prior to initiation of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 274 cartridge cases, 254 coated and 20 uncoated for performance of examination and tests as specified in 4.3 utilizing the test procedures of 4.4 The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used for regular production. All parts and materials, including packaging and packing must be the same as will be used for regular production and shall be obtained from the same source of supply. The supplier shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all the requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample has been submitted. Two hundred and fifty of the above-coated cartridge cases shall be forwarded to a designated proving ground for ballistic testing.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The ordnance standard finish blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Case, Cartridge, Drawing F7548025 - prior to coating.

	Defect	Method of Inspection
CRITICAL		
1.	Cracks	Visual
MAJOR		
101.	Concentricity of primer hole with body basic diameter	Gage
102.	Concentricity of primer counterbore with primer hole	Gage
103.	Diameter of flange, min.	Gage
104.	Diameter of mouth	Gage
105.	Diameter of primer hole	Gage
106.	Depth of primer hole counterbore, min.	Gage
107.	Length from primer hole counterbore to forward surface of flange, min.	Gage
108.	Taper in primer hole	Gage
109.	Thickness of wall at mouth	Gage
110.	Thickness of wall at basic length near mid-body	Gage
111.	Thickness of wall at basic length near head	Gage
112.	Thickness of flange, min.	Gage
113.	Thickness of head, min.	Gage
114.	Contour of inside of head and sidewalls not smooth	Visual
MINOR		
201.	Diameter of primer hole counterbore	Gage
202.	Diameter of body under flange, min.	Gage
203.	Diameter of forward obturating bead, min.	Gage
204.	Diameter of rear obturating bead, min.	Gage
205.	Length to forward obturating bead	Gage
206.	Length to rear obturating bead	Gage

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4.3.2.1 (Cont.) MINOR

	Defect	Method of Inspection
207.	Length to basic diameter near mouth, min.	Gage
208.	Length to basic diameter near mid-body, min.	Gage
209.	Length to basic diameter near head, min.	Gage
210.	Length total	Gage
211.	Radii or chamfers, missing or incorrect	Visual (see 6.2)
212.	Surface finish, improper	Visual (see 6.2)
213.	Burr	Visual
214.	Foreign matter	Visual

4.3.2.2 Case, Cartridge, Drawing F7548025 - after coating.

CRITICAL
None defined.

MAJOR

101.	Metal defective (unless otherwise classified)	Visual
102.	Protective coating inadequate or damaged	Visual

MINOR

201.	Dent or damaged	Visual
202.	Varnish in primer hole counterbore	Visual
203.	Varnish in primer hole	Visual
204.	Silicone compound missing from primer hole or primer hole counterbore	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Foreign matter	Visual

4.3.2.3 Carton, Drawing F7548235 - prior to sealing.

CRITICAL
None defined.

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4.3.2.3 (Cont.)

Defect	Method of Inspection
MAJOR	
None defined.	
MINOR	
201. Improper assembly	Visual
202. Incorrect number of cartridge cases in carton	Visual
203. Separator missing or damaged	Visual

4.3.2.4 Carton, Drawing F7548235 - after sealing.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton improperly sealed	Visual
203. Contents loose	Visual and Manual
204. Marking missing, incorrect or illegible	Visual

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4.3.3 Maximum profile (major defect).- Each cartridge case shall be gaged in a Government certified maximum profile gage. Failure of the cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the cartridge case.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.4.1 Mechanical properties

4.3.4.1.1 Hardness.- Twenty cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of the applicable drawing shall be cause for rejection of the sample or lot.

4 3.4.1.1.1 Thermal treatment (major defect).- Each cartridge case shall be subjected to a hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.4.1 or indications that the cartridge case failed to harden shall be cause for rejection of the cartridge case.

4.3.4.1.2 Yield strength, tensile strength and elongation.- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. The results shall be recorded for information only. The cartridge cases used for the hardness test may be used for this test.

4.3.4.2 Marco examination.- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be examined for folds and cold shuts. Failure of one or more cartridge cases to comply with 3.9.1.1 shall be cause for rejection of the sample or lot. The cartridge cases used for the hardness test may be used for this test.

4.3.4.3 Protective coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.4.4 Functional.- The following functional tests will be performed by the Government.

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4.3.4.4.1 Initial production sample.- The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table II, except no retest will be premitted.

Table I

Test Phase	Quantity to be Tested
Excess Pressure Phase	100
Cold Temperature Phase	50
Excess, Excess Pressure Phase (for Information Only)	100

4.3.4.4.2 Production lot sample.- One hundred cartridge cases from each production lot shall be ballistically tested in the excess pressure phase. Acceptance and rejection criteria shall be in accordance with Table II. The lot shall be considered eligible for retest using double the number of samples provided that the rejection cirteria of Table II has not been attained. A retest will not be permitted if a split or crack occurs within one caliber of the head in one or more cartridge cases.

Table II

Defect Characteristic	First Sample		Retest Cumulative Sample	
	Accept	Reject	Accept	Reject
1. Obturation	0	2	1	2
2. Splits or cracks with 1 Caliber of the Head	0	1	Not Permitted	
3. Splits or cracks in sidewall more than 1 Caliber from the Head	1	3	2	3
4. Extraction	0	2	1	2
5 Primer Retention	0	2	1	2

4.3.5 Inspection equipment.- Index of Inspection Equipment Lists Number IEL-7548025 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL (reference is made to 6.4.1 for definition of coding format).

4.4 Test methods and procedures

4.4.1 Mechanical properties

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4.4.1.1 Hardness.- This test shall be performed on locations specified by the applicable drawing in accordance with the procedures prescribed for Rockwell Hardness Testing by Federal Test Method Standard No. 151.

4.4.1.1.1 Thermal treatment.- This test shall be perfromed on a hardness comparator of a non-destructive type approved by the Contracting Officer.

4.4.1.2 Yield strength, tensile strength and elongation.- The test specimens shall be taken from the cartridge case in accorancee with the standard position and size of tensile specimen chart, Figure I, the yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed for these tests by Federal Test Method Standard No. 151.

4.4.2 Protective coating.- The samples provided by 4.3.4.3 shall be tested using the procedures and equipment prescribed by the applicable specification.

4.4.3 Maximum profile gage.- The cartridge case, after application of protective coating, shall freely enter the Government certified maximum profile gage. The gate of the gage shall close without interference with cartridge case head.

4.4.4 Marco examination.- The cartridge case shall be cut in 90 degree section along the axis through the primer hole. The specimen shall be etched in accordance with the procedures prescribed for this test by Federal Test Method Standard No. 151.

4.4.5 Functional

4.4.5.1 Initial production sample

4.4.5.1.1 Excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit (F) to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.2 Cold temperature phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 36,400 pounds per square inch (PSI) to 39,700 (PSI), Piezo equivalent pressure, when temperature conditioned at 70 degrees F. The complete round will be temperature conditioned at minus 35 degrees F., to minus 45 degrees F , and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

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4.4.5.1.3 Excess, excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.2 Production lot

4.4.5.2.1 Excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A - Not applicable

5.1.2 Level B - Not applicable

5.1.3 Level C - Cartridge cases shall be packed in accordance with the requirements of Drawing F7548235.

5.2 Labeling and marking.- Labeling and marking for shipment shall be in accordance with Specification MIL-P-10025.

5.3 Data card.- Data card information shall be as specified in Publication QAI-14-210-1, Preparation of Ammunition Data Cards.

6. NOTES

6.1 Prior to fabrication of an initial production sample, the supplier, upon approval of the Contracting Officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The supplier shall furnish all information pertaining to the manufacture of the lots to the Contracting Officer. All costs for testing these process control lots will be borne by the Government.

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6.2 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Extraction.-The term "Extraction" shall mean that the fired cartridge case is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

6.4 Inspection equipment lists.- Inspection equipment lists (IEL) indicate availability of inspection equipment designs by showing in the "Number" column of the list of inspection equipment (OO Form 1243-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.4.1.

6.4.1 IEL Codes:

(AD) Army design.- Army designs are detailed drawings which depict completely all the information necessary for the fabrication of the item of inspection equipment. The supplier need provide no design when an Army design is listed for an item of inspection equipment.

(SCD) Specification control drawing.- Specification control drawings depict the minimum equipment requirements in outline. Descriptive, diagrammatic, or pictorial form specifies required performance or other characteristics. Such drawings are provided by the Government as an attachment to the pertinent inspection equipment lists. Approval of items of inspection equipment fabricated in accordance with these drawings is delegated to the inspection element of the Army Procurement District administering the contract.

(CE) Commercial equipment.- Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and the Government. Suppliers are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the Army Procurement District administering the contract.

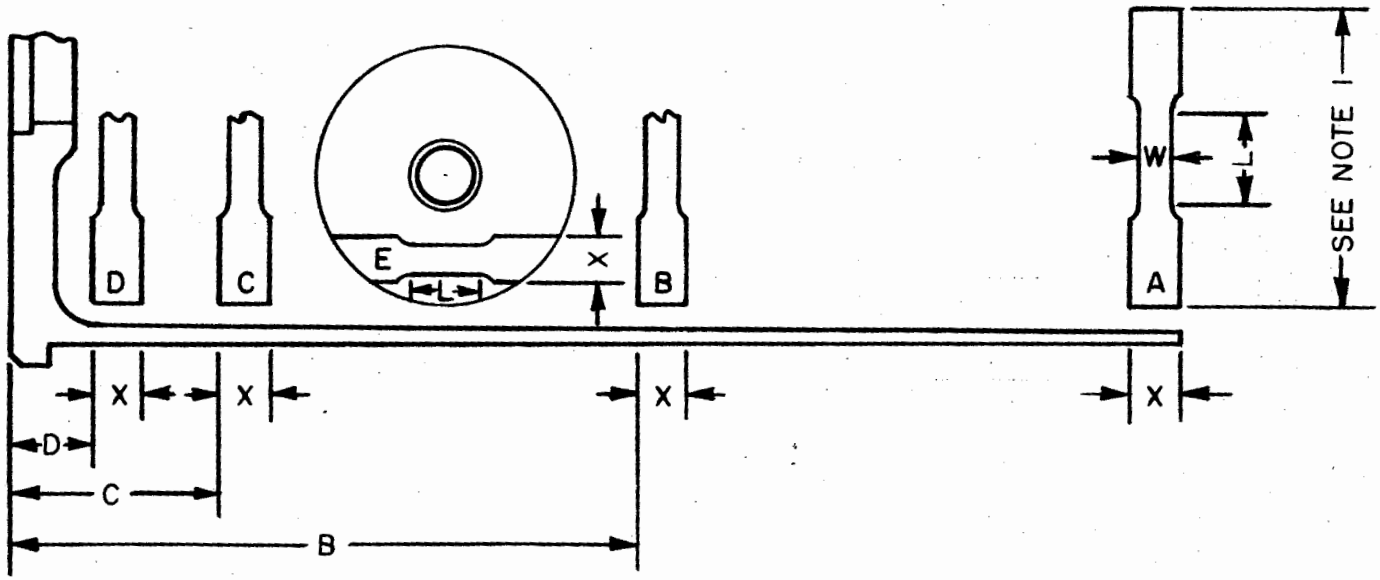
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(CD) Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Such designs shall be submitted for approval to the Commanding Officer Frankford Arsenal, Attn: SMUFA-Q3200, Philadelphia, Pa. 19137.

Custodian:
Army - MU

Preparing activity:
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Project No. 1315 - A429

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SPECIMEN	X	W	L	LOCATION FROM HEAD
A	.40	.25	1.0	At Mouth
B	.40	.25	1.0	7.50
C	.40	.25	1.0	2.50
D	.40	.25	1.0	1.00
E	.40	.25	1.0	In Base

NOTES:

1 - Overall length of specimen to suit fixture in testing machine.

2 - Specimens C and D to be machined to uniform thickness parallel to inside surface.

3 - Specimen E to be taken from center of head midway between primer hole counterbore and edge of flange.

4 - Specimen E to be machined so as to eliminate effects of stamping on test.

5 - Yield strength in 0.2 percent offset

6 - Percent elongation in 1.0 inch gage length

STANDARD POSITIONS AND SIZES OF
TENSILE SPECIMENS FOR CASE, CARTRIDGE, 105MM, M14B1

FIGURE I

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